

"HRV-Based Mortality Prediction at 4, 8, and 12 Hours Before Death in Ischemic Stroke Patients"

2-4 h			
Variables	Survived(194)	non-Survived(34)	P value
Demograaphics			
Age,y	69.73±13.09	71.26±12.59	0.523
Sex,men,n(%)	95(48.9%)	17(50%)	1.000
HRV variables			
HRV_MeanNN	788.33±176.18	624.75±148.34	<0.001
rrIntv_percentage75	0.83±0.20	0.66±0.16	<0.001
HRV_Prc80NN	846.89±237.09	668.62±158.63	<0.001
HRV_MedianNN	775.42±153.77	616.29±152.49	<0.001
rrIntv_percentage25	0.73±0.16	0.58±0.16	<0.001
HRV_Prc20NN	723.39±156.98	575.81±156.03	<0.001
HRV_SDRMSSD	1.31±0.63	1.05±0.32	<0.001
HRV_CSI	2.34±1.36	1.79±0.71	<0.001
HRV_SDANN1	32.25±82.98	12.38±9.85	0.0015
rrIntv_min	0.59±0.15	0.50±0.16	0.0022

6-8 h			
Variables	Survived(208)	non-Survived(34)	P value
Demograaphics			
Age,y	70.20±13.04	71.26±12.59	0.655
Sex,men,n(%)	101(48.6%)	17(50%)	1.000
HRV variables			
HRV_MeanNN	804.14±190.33	630.77±142.08	<0.001
rrIntv_percentage75	0.85±0.21	0.67±0.15	<0.001
HRV_Prc80NN	863.15±236.57	678.50±156.87	<0.001
HRV_MedianNN	791.85±165.13	622.70±149.29	<0.001
rrIntv_percentage25	0.75±0.17	0.59±0.15	<0.001
HRV_Prc20NN	736.62±168.46	582.97±149.49	<0.001
HRV_SDRMSSD	1.29±0.65	0.99±0.31	<0.001
HRV_TINN	168.48±129.84	86.68±96.55	<0.001
HRV_CSI	2.29±1.39	1.65±0.69	<0.001
HRV_CSI_Modified	860.32±1363.84	408.22±382.27	<0.001
HRV_MFDFA_alpha2_Mean	0.92±0.30	0.75±0.22	<0.001
HRV_HFD	1.85±0.15	1.92±0.10	<0.001
rrIntv_max	1.22±1.18	0.87±0.33	<0.001
HRV_MaxNN	1224.57±1179.91	867.44±331.09	<0.001
HRV_MFDFA_alpha2_increme nt	0.10±0.13	0.05±0.06	<0.001

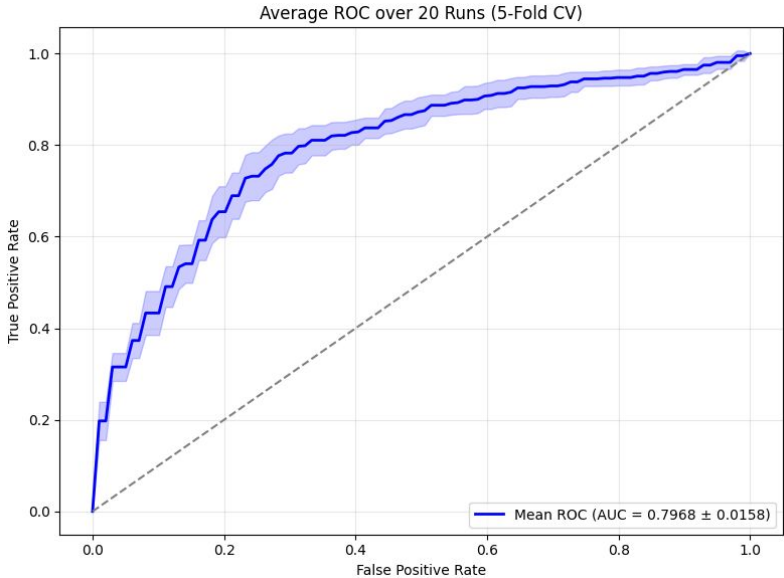
10-12 h			
Variables	Survived(199)	non-Survived(31)	P value
Demograaphics			
Age,y	71.40±11.84	71.90±11.97	0.832
Sex,men,n(%)	96(48.2%)	15(48.3%)	1.000
HRV variables			
HRV_TINN	166.48±129.26	91.00±95.06	<0.001
HRV_MFDFA_alpha2_Fluctuation	0.002±0.004	0.001±0.001	<0.001
rrIntv_percentage25	0.74±0.16	0.63±0.18	0.0012
HRV_Prc20NN	731.41±157.95	618.91±178.10	0.0020
HRV_MFDFA_alpha2_Mean	0.91±0.30	0.72±0.29	0.0024
rrIntv_MaxHistCounts	66.36±58.17	126.37±105.55	0.0040
HRV_SDRMSSD	1.27±0.61	1.02±0.39	0.0041
HRV_CSI	2.25±1.33	1.71±0.89	0.0049

Threshold Comparison
at Sensitivity Levels 0.95, 0.90, and 0.85
in Logistic Regression(Average of 20 runs of 5-fold cross-validation)

2-4h LR (10 HRV features)

Threshold	Sensitbity	Specificity	Accuracy	Precision	F1 score	Auc
0.124	0.95	0.1858	0.2998	0.1704	0.2888	0.7968
0.258	0.90	0.3843	0.4612	0.206	0.3348	0.7968
0.383	0.85	0.542	0.5879	0.2464	0.3818	0.7968

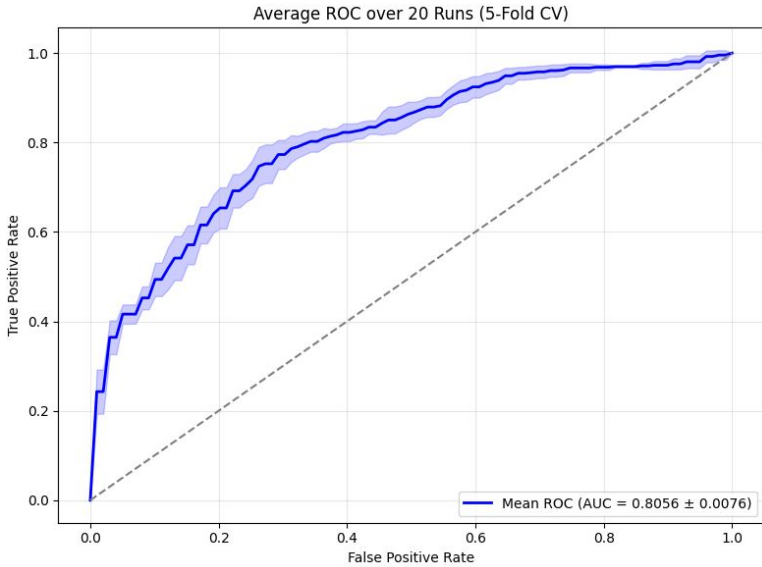
Variables	Coefficient	Odds ratio
rrIntv_min	0.6182	1.8555
HRV_CSI	-0.1019	0.9031
HRV_Prc20NN	-0.2174	0.8046
rrIntv_percentage25	-0.2313	0.7935
HRV_SDANN1	-0.2988	0.7417
HRV_MeanNN	-0.3088	0.7344
HRV_Prc80NN	-0.3546	0.7015
rrIntv_percentage75	-0.4191	0.6576
HRV_SDRMSSD	-0.4394	0.6444
HRV_MedianNN	-0.5051	0.6035



6-8h LR (15 HRV features)

Variables	Coefficient	Odds ratio
HRV_CSI_Modified	0.4915	1.6348
rrIntv_percentage25	-0.0236	0.9767
HRV_Prc20NN	-0.0640	0.9379
HRV_MFDFA_alpha2_Mean	-0.0938	0.9105
HRV_Prc80NN	-0.1465	0.8637
rrIntv_percentage75	-0.2281	0.7961
HRV_MeanNN	-0.2521	0.7777
HRV_MaxNN	-0.2852	0.7519
rrIntv_max	-0.2852	0.7519
HRV_MedianNN	-0.3026	0.7389
HRV_HFD	-0.3249	0.7226
HRV_CSI	-0.3692	0.6913
HRV_SDRMSSD	-0.5110	0.5999
HRV_MFDFA_alpha2_increment	-0.6843	0.5045
HRV_TINN	-0.8049	0.4471

Thredhold	Sensitbity	Specificity	Accuracy	Precision	F1 score	Auc
0.28	0.95	0.2875	0.3806	0.1791	0.3013	0.8056
0.356	0.90	0.4404	0.505	0.2082	0.3381	0.8056
0.398	0.85	0.5029	0.5504	0.2185	0.3476	0.8056



10-12h LR (8 HRV features)

Thredhold	Sensitbity	Specificity	Accuracy	Precision	F1 score	Auc
0.153	0.95	0.2327	0.3293	0.1617	0.2764	0.7534
0.228	0.90	0.3399	0.4154	0.1753	0.2934	0.7534
0.287	0.8516	0.4183	0.4767	0.1858	0.305	0.7534

Variables	Coefficient	Odds ratio
rrIntv_MaxHistCounts	0.4145	1.5136
HRV_Prc20NN	-0.1293	0.8787
HRV_CSI	-0.1367	0.8722
HRV_SDRMSSD	-0.1529	0.8581
rrIntv_percentage25	-0.1760	0.8386
HRV_MFDFA_alpha2_Mean	-0.2960	0.7438
HRV_TINN	-0.5149	0.5975
HRV_MFDFA_alpha2_Fluctuation	-1.1455	0.3181

